SILVERZONE

Do not start attempting the test paper until you are asked to do so.

ENROLLMENT NUMBER:

STUDENT NAME:

Class 6

Maximum Marks : 100
Duration : 40 Minutes
Total Questions : 35

Test Paper Type
**AD**

Level - 2

# iOS 14th International Olympiad of Science 2022-2023

## Instructions for the Candidates

1. Write your 12 digit iOS Enrollment Number and your name on top of the Question Paper in the given space.
2. Filling up of improper Enrollment Number may lead to unavailability of Result.
3. This Question Paper cum Answer Sheet must be submitted to your teacher after the completion of the test.
4. Tick ☑ your answer (A, B, C or D) on the given box ☐ in this Question Paper cum Answer Sheet with HB Pencil or Black/Blue Ball point Pen.
5. The Question Paper cum Answer Sheet Booklet consists of 35 questions, divided into three sections.
6. Section-A: Science (23 questions); Section-B: Reasoning and Aptitude (7 questions); Section-C: Scholar's Zone (5 questions).
7. Each question of Section-A carries 2.5 marks; Section-B carries 3 marks and that of Scholar's Zone carries 4.3 marks.
8. Use of calculator or any other devices in the examination is strictly prohibited.
9. Rough work should be done on the blank sheet/space provided in the question paper booklet.
10. All the questions are compulsory and there is no negative marking.

SIGN./NAME OF THE STUDENT

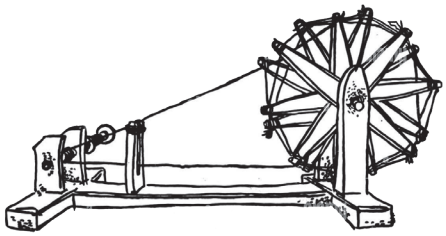
SIGN. OF THE TEACHER

1. The symptoms shown below is of which disease?
   *1. Weak muscles*
   *2. Very little energy to work*
   A. ☐ Night blindness B. ☐ Beriberi
   C. ☐ Scurvy D. ☐ Goitre

2. X is a natural fibre. It is obtained from the stem of a plant. X is a rough fibre, so it is not used for making fabrics however it is used to make ropes and gunny bags. Identify X.
   A. ☐ Jute fibre B. ☐ Cotton fibre
   C. ☐ Wool fibre D. ☐ Nylon fibre

3. Which organs of a bird are modified as wings?
   A. ☐ Forelimbs B. ☐ Hindlimbs
   C. ☐ Skin D. ☐ Tail

4. The device shown below is used to make:



   A. ☐ fabrics from yarns
   B. ☐ yarns from fibres
   C. ☐ fibres from yarns
   D. ☐ yarns from fabrics

5. Which one of the following materials is a poor conductor of electricity?
   A. ☐ Wood B. ☐ Graphite
   C. ☐ Copper D. ☐ Aluminium

6. The property used, in the method of separation shown below, is:

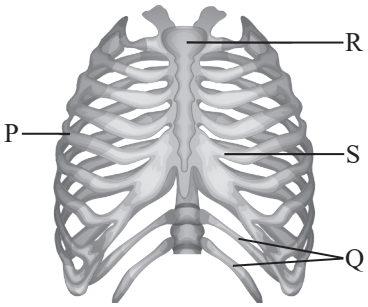


   A. ☐ difference in size
   B. ☐ difference in weight
   C. ☐ difference in colour
   D. ☐ difference in texture

7. Which among the following is the natural magnet?
   A. ☐ Steel
   B. ☐ Cobalt
   C. ☐ Magnetite
   D ☐ Ebonies

8. Which of the following changes cannot be reversed?
   A. ☐ Increase in temperature of water
   B. ☐ Bending of an iron rod
   C. ☐ Turning milk sour
   D. ☐ Dissolution of salt into water

9. Which letter represents floating ribs in the figure shown below?




   A. ☐ P B. ☐ Q
   C. ☐ R D. ☐ S

10. Motion of a child on a swing is an example of:
    A. ☐ rectilinear motion
    B. ☐ periodic motion
    C. ☐ rotational motion
    D. ☐ circular motion

11. For which of the following needs does an organism depends on its habitat?
    A. ☐ Food
    B. ☐ Water
    C. ☐ Shelter
    D. ☐ All of these

12. Which type of worms are used in vermicomposting?
    A. ☐ Redworms B. ☐ Liver fluke
    C. ☐ Guinea worm D. ☐ All the above

13. Which one of the following statements about food is incorrect?
    A. ☐ Food is a substance that provides us energy for all the activities that we do.
    B. ☐ Plants and animals are the main sources of our food.
    C. ☐ Food protects us from various diseases.
    D. ☐ Food is not essential for plants.

14. What is the thread like structure of a climber plant called which coils around any neighbouring object and helps the plant climb up?

A. ☐ Root
B. ☐ Lamina
C. ☐ Tendril
D. ☐ Vascular system

15. Which one of the following combinations of features would you observe in the plant given below?



A. ☐ Parallel venation and tap root
B. ☐ Parallel venation and fibrous root
C. ☐ Reticulate venation and tap root
D. ☐ Reticulate venation and fibrous root

16. Switch is a device which is used to:

A. ☐ open or close an electric circuit.
B. ☐ measure the current flowing in an electric circuit.
C. ☐ measure the voltage of battery.
D. ☐ protect the electrical appliances from current.

17. Which one of the following is not a unit to measure length?

A. ☐ m
B. ☐ cm
C. ☐ mg
D. ☐ mm

18. Which of the following is a magnetic material?

A. ☐ Silver
B. ☐ Plastic
C. ☐ Cobalt
D. ☐ Glass

19. Which one of the following pairs of materials is insoluble in water?

A. ☐ Groundnut oil and coconut oil
B. ☐ Milk and salt
C. ☐ Sand and alum
D. ☐ Chalk powder and sugar

Space for Rough Work

20. Which one of the following statements is incorrect about air?

A. ☐ Air is a colourless substance
B. ☐ Air can be compressed
C. ☐ Air dissolves in water
D. ☐ Air does not occupy space

21. Which process plays an important role in bringing water from the atmosphere back to the surface of the earth?

A. ☐ Melting
B. ☐ Evaporation
C. ☐ Condensation
D. ☐ Transpiration

22. Which gas is essential for burning to occur?

A. ☐ Nitrogen
B. ☐ Carbon dioxide
C. ☐ Hydrogen
D. ☐ Oxygen

23. Shadow is formed when:

A. ☐ colour of light becomes black
B. ☐ colour of an object becomes black
C. ☐ an opaque object obstructs the path of light
D. ☐ light comes to rest

Space for Rough Work

24. Let a * b * c * d * e = ab – de + c.
If x = 7 * 6 * 5 * 4 * 3,
then what is the value of x * 11 * 9 * 8 * 7?

A. ☐ 320 B. ☐ 329
C. ☐ 338 D. ☐ 348

25. Find the missing number from the given alternatives.

| | | |
|---|---|---|
| 24 | 144 | 384 |
| 6 | 36 | ? |
| 2 | 12 | 32 |
| 1 | 6 | 16 |

A. ☐ 91 B. ☐ 86
C. ☐ 80 D. ☐ 96

26. Find the wrong number in the given series?
10, 20, 34, 52, 72, 100, 130

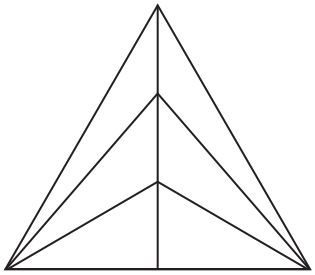
A. ☐ 130 B. ☐ 100
C. ☐ 72 D. ☐ 52

27. A person starts from his house and moves towards the market. He walks 40 m towards South and takes a right turn. After walking 30 m he takes a left turn and walks 20 m. Finally, he takes a left turn and reaches the market after walking 30 m. How far and in which direction is his house from the market?

A. ☐ 60 m, East B. ☐ 80 m, North
C. ☐ 60 m, South D. ☐ None of these

Space for Rough Work

28. How many triangles are given in the figure below?



A. ☐ Less than 10 B. ☐ Between 12 to 18

C. ☐ 11 D. ☐ More than 18

29. Using the table given below, identify the correct expression.

| P | Q | R | S |
|---|---|---|---|
| × | – | + | ÷ |

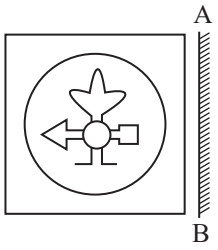
A. ☐ 18 P 6 R 7 S 5 Q 2 = 20

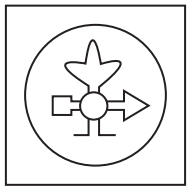
B. ☐ 18 S 6 P 7 Q 5 R 2 = 18

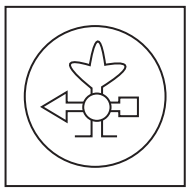
C. ☐ 18 Q 6 S 7 P 5 R 2 = 16

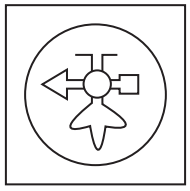
D. ☐ 18 P 6 Q 7 S 5 R 2 = 22

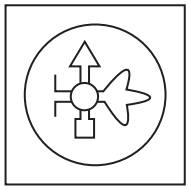
30. Which of the answer figures is exactly the mirror image of the question figure, when the mirror is held on the line AB?




A. ☐ 

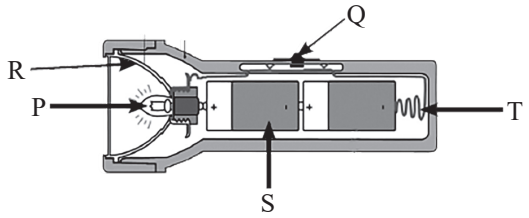
B. ☐ 

C. ☐ 

D. ☐ 

Space for Rough Work

31. Label the electric torch with the suitable words.




A. ☐ P → Bulb, Q → Slide switch, R → Metal spring, S → Electric cell, T → Reflector

B. ☐ P → Slide switch, Q → Bulb, R → Reflector, S → Electric cell, T → Metal spring

C. ☐ P → Metal spring, Q → Slide switch, R → Bulb, S → Electric cell, T → Reflector

D. ☐ P → Bulb, Q → Slide switch, R → Reflector, S → Electric cell, T → Metal spring

32. Which of the following statement(s) is/are correct?

*i.* *The solid particles which are insoluble in a liquid can be separated by decantation.*

*ii.* *A mixture of silver filings and sulphur powder can be separated by using a magnet.*

*iii.* *A mixture contains molecules of two or more substances.*

*iv.* *Soluble solids can be separated from liquid by evaporation.*

A. ☐ only i

B. ☐ only i and iii

C. ☐ only i, iii and iv

D. ☐ All i, ii, iii and iv

Space for Rough Work

33. Fill in the blanks with the correct combination of words.

*The* p *of earth around the* q *is an example of* r *motion.*

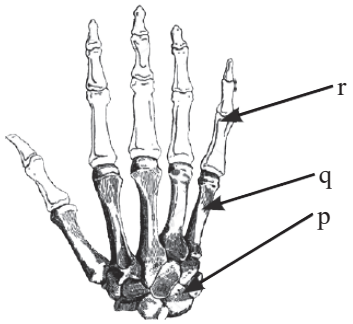
A. ☐ p → rotation, q → moon, r → periodic
B. ☐ p → revolution, q → jupiter, r → rotational
C. ☐ p → rotation, q → axis, r → rectilinear
D. ☐ p → revolution, q → sun, r → periodic

34. If you happen to go to a desert, what changes do you expect to observe in the urine you excrete? You would:

*i. excrete small amount of urine.*
*ii. excrete large amount of urine.*
*iii. excrete concentrated urine.*
*iv. excrete very dilute urine.*

A. ☐ i and iii
B. ☐ ii and iv
C. ☐ i and iv
D. ☐ i and ii

35. Look at the following figure and choose the correct option.




A. ☐ p → phalanges, q → carpals, r → metacarpals
B. ☐ p → metacarpals, q → carpals, r → phallanges
C. ☐ p → carpals, q → metacarpals, r → phallanges
D. ☐ p → carpals, q → phallanges, r → metacarpals

Space for Rough Work

ANSWERS

INTERNATIONAL OLYMPIAD OF SCIENCE 2022 (LEVEL 2) - CLASS_6

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | (B) | 2. | (A) | 3. | (A) | 4. | (B) | 5. | (A) | 6. | (B) | 7. | (C) | 8. | (C) | 9. | (B) | 10. | (B) |
| 11. | (D) | 12. | (A) | 13. | (D) | 14. | (C) | 15. | (B) | 16. | (A) | 17. | (C) | 18. | (C) | 19. | (A) | 20. | (D) |
| 21. | (C) | 22. | (D) | 23. | (C) | 24. | (C) | 25. | (D) | 26. | (C) | 27. | (D) | 28. | (B) | 29. | (B) | 30. | (A) |
| 31. | (D) | 32. | (C) | 33. | (D) | 34. | (A) | 35. | (C) | | | | | | | | | | |